Scale, proportions, circles, and review

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

6. Mix A has 2 cups juice and 3 cups water. Mix B has 3 cups juice and 5 cups water. Which has more juice for the same water?

7. Find $3\frac{1}{2} \div \frac{3}{4}$.

8. Solve $1.5x = 4.80$.

9. Find $|0|$ and describe it as a distance.

10. Find the median of 8, 2, 4, 6.

11. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

12. A circle has radius 2 cm. Find circumference exactly and using 3.14.

13. A circle has radius 2 cm. Find area exactly and using 3.14.

14. A circle has diameter 4 cm. Find area exactly and using 3.14.

15. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

16. A rectangle 6 cm by 5 cm is scaled by 2. Find new dimensions and area.

17. A drawing has area 30 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

18. A map uses 1 cm for 6 km. A route is 5.5 cm on the map. Find actual distance.

CHALLENGE / Explain your plan and check the result

C1. A circular garden has radius 7 m. A2 m gate replaces part of the fence. Using$\pi \approx 3.14$, find the fence length and the cost at $5 per meter. Explain why subtracting the gate from the area would be wrong.

C2. A jacket costs $80. Store A reduces the price by 25%, then adds 10% tax. Store B adds 10% tax first, then reduces the taxed total by 25%. Find both final prices. Explain why these two orders agree.